Synthesis of Chinese *Dizi* Ornaments

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Abstract

The *dizi* is a Chinese transverse flute that produces a characteristic nasal buzzing tone. This project uses a line segment method for performing grace notes. To model a trill, we used one function table for frequency modulation and another for amplitude modulation. We combined the two designs to connect grace notes to a trill. This method produces realistic *dizi* ornaments that sound better than overlapping the tones and are easier to use than connecting them all with a line segment.

1 Introduction

The *dizi* is usually made of bamboo, with a cane membrane over one hole (see Figure 1) that produces a characteristic nasal buzzing tone rich in upper partials. Varying numbers of grace notes often precede or follow *dizi* trills. This project uses a combination of a line segment with frequency and amplitude modulation functions to make realistic *dizi* ornaments.

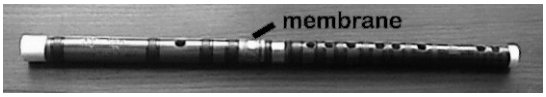


Figure 1. Dizi.

We studied the musical characteristics of *dizi* ornaments to answer the questions: What is the most accurate and efficient method to synthesize those characteristics using the group wavetable synthesis instrument design from our previous work? Is the method good enough to fool listeners into believing that humans played the synthesized examples?

After comparing the results, we concluded that the line segment method is an efficient method for modeling accurate slurred grace notes, but trills sound better using separate function tables for frequency and amplitude modulation, and the new method is easier to use. Combining the two methods is highly successful and the ornaments sound realistic enough to fool listeners into believing that humans played our synthesized examples.

2 Background

Previous research modeled the spectra of ordinary single tones of many wind instruments [1][2][3][4][5]. We previously synthesized individual *dizi* tones using a wavetable model that is easy to use and sounds like the original tones [6][7]. We used the same design to model the *dizi* tones in this project. Isolated synthesized tones may be indistinguishable from acoustic tones on listening tests, but musicians don't always play isolated tones! They connect tones to make "musical" performances. Since wavetable synthesis is straightforward and matching works well for isolated *dizi* tones, we will focus our discussion on modeling ornaments connected to the tones, such as grace notes and trills.

3 Synthesizing Grace Notes

Overlapping can simulate slurs when the decay of one tone can overlap the attack of the following tone. But this technique doesn't work on grace notes because the notes are too short, and they can even have clicks. Connecting together very short tones, therefore, often makes it obvious that the music is synthesized. Connecting the frequencies of two notes with a line segment and morphing the transition using parameter interpolation [8] obviously can give a smoother slur.

We created an instrument which cross-fades two unison signals sharing one frequency line segment and phase, but using their correct wavetables (see Figure 2). Amplitude envelopes cross-fade the two signals as with overlapped notes, but it is smoother because the frequency change in the transition more closely resembles one on a real *dizi*, and using the same phase (and the other frequency parameters, such as noise and vibrato) also makes the cross-fading itself less noticeable. We also increased the noise in the cross-fade as the amplitude decreases, so that the maximum amount of noise is in the middle of the cross-fade, at the point of minimum amplitude.

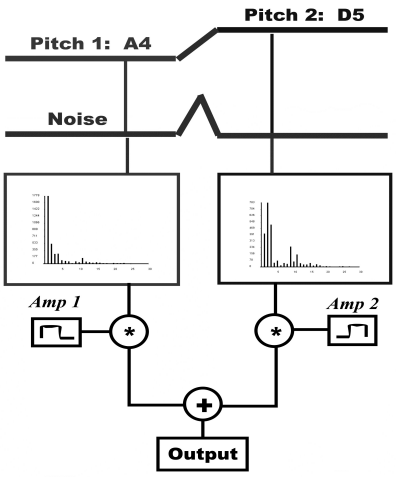


Figure 2. Cross-Fading Two Notes with Noise in the Transition.

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We can use the same method to play additional notes. Our design can play three slurred notes, allowing one grace note before and one after the base note.

4 Synthesizing Trills

But how can we slur varying numbers of notes in trilled phrases? In Csound, slurred groups of different numbers of notes need different numbers of input parameters. Requiring the maximum number of input parameters for every note statement and using placeholder numbers for those we don't need results in an extremely cumbersome design. A whole note trilled quickly in a slow tempo can easily contain more than 32 individual notes! Can a repeating function controlling the frequency of a single tone produce smoother trilled notes than overlapping tones does?

5 Trill Frequency Modulator

What is the average repeating pattern that best represents the basic frequency change for this trill as well as others? The function does not need to model pitch variation of the average tone, change of speed or jitter, so it can represent one average cycle of the trill, and adjusting the parameters randomly within their typical ranges can vary each cycle.

The frequency modulator must use a function that can approximate the frequency analysis graph of the trill shown in Figure 3.

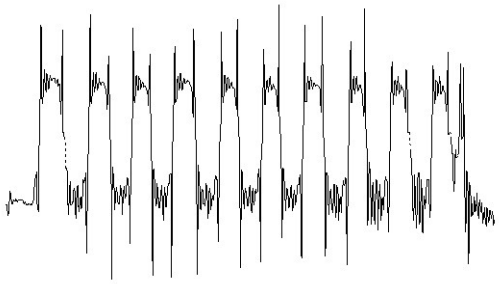


Figure 3. Frequency Analysis of an A5 to B5 Trill.

Figure 4 shows a close-up of one cycle of this trill. Averaging one cycle provides a good shape for a trill frequency function.

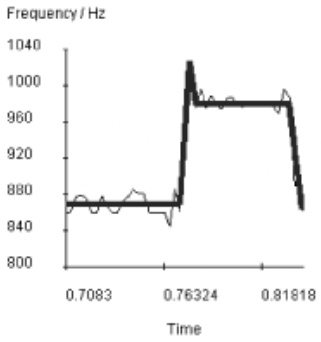


Figure 4. Averaging One Cycle of the Trill for a Frequency Function.

The function must oscillate between the frequency of the lower note and the frequency of the higher note. The cycles begin with a slight overshooting of the required frequency, perhaps 20% (see Figure 5), and we randomly vary the trill rate.

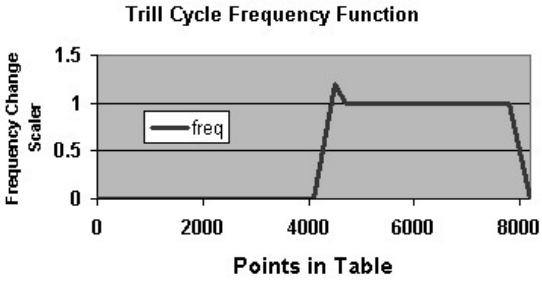


Figure 5. Trill Cycle Frequency Function.

6 Trill Amplitude Modulator

Figure 6 shows the waveform of two cycles of the trill. The waveform shows a drop-off in amplitude in the transitions between the trilled notes. It also shows that one of the notes is a little bit louder than the other.

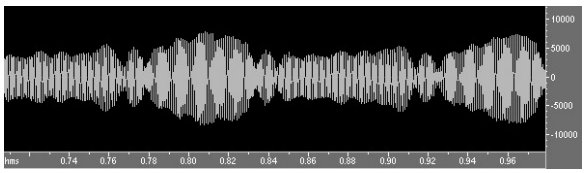


Figure 6. Two Cycles of Waveform of Dizi A4 to B4 Trill.

What is the optimal shape for the trill amplitude function? Analyzing the two trill cycles shown in Figure 6 provides more detail on the sharp amplitude drops during the transitions between the notes (see Figure 7). In addition, the amplitude shows an extra spike at the beginning and ending of the period.

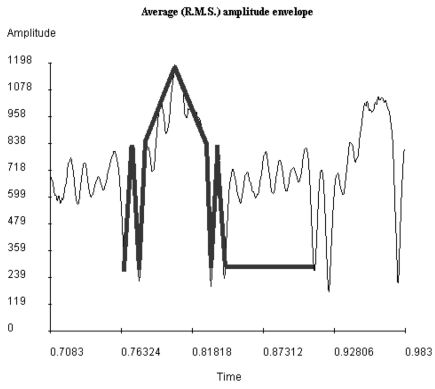


Figure 7. Averaging a Cycle of the Trill for an Amplitude Function.

To get a realistic trill, we must model the amplitude changes of the transition. Although the most important amplitude changes occur at the same time as the most important frequency changes, we cannot use the same function to control both the frequency and amplitude because the frequency peak is at the beginning of the cycle and the amplitude peak is in the middle. In addition, when notes in tremolos alternate by leap, each note must use its own wavetable for its spectrum. This requires cross-fading two signals while the first function alternates their frequencies.

The average amplitude of a trill cycle provides a separate function that must oscillate between the amplitude of the lower note and the amplitude of the higher note (see Figure 8). The cycles begin and end with a slight amplitude spike during the transition between the two notes.

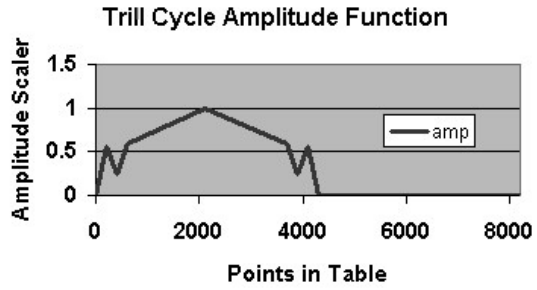


Figure 8. Trill Cycle Amplitude Function.

We stored the new trill shape in a function table and then used an amplitude modulator in that shape to alternately fade the amplitudes of the two signals in and out (see Figure 9).

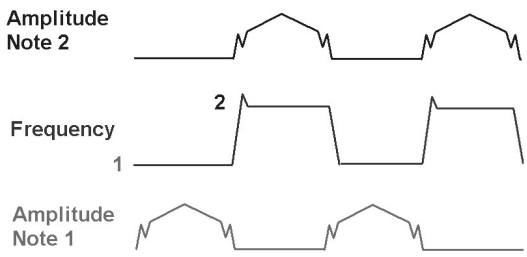


Figure 9. Combining the Functions to Produce a Trill.

This design permits several score parameters to control the trill rate using a line segment. For example, our design uses four score parameters to control an initial, middle and final tremolo rate, and the time required to change from the first tremolo rate to the second. A separate score parameter controls the changing amplitude of the trill as it would for a single sustained tone. This method is much easier than typing many notes and trying to manually adjust their start times, durations and amplitudes to get a naturally-varying rate and amplitude quality.

7 Combining Ornaments with the Trill

Trilled notes on the *dizi* may be preceded or followed by grace notes. A four-note design covers many of the possibilities (see Figure 10).

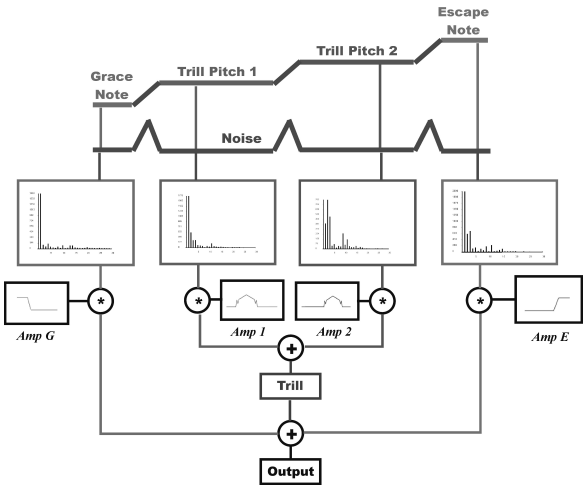


Figure 10. Summary of the Four-Note Design.

We assigned one note for the preceding grace note and one for the escaping grace note, with a trilled pair of notes between the grace note and the escape note. A single line segment, shared among all the notes, controls their overall frequency (see Figure 11). The tremolo rate is set to 0 during the grace notes, since they are not part of the tremolo, and it varies as previously described during the tremolo between the grace notes. The amplitude envelope of each signal is set to 0 during the time it is not active.

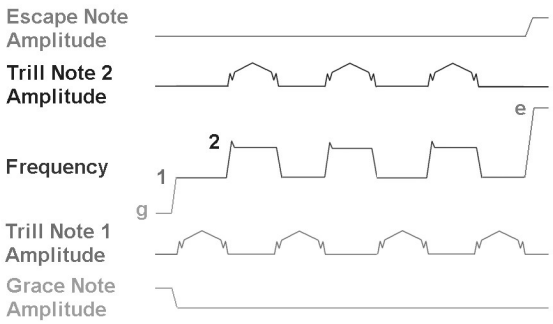


Figure 11. Adding Grace Notes Before and After the Trill.

8 Conclusion

After comparing the results, we concluded that the line segment method is an efficient method for modeling accurate slurred grace notes, but trills sound better using separate function tables for frequency and amplitude modulation, and the new method is easier to use. Combining the two methods

is highly successful and the ornaments sound realistic enough to fool listeners into believing that humans played our synthesized examples.

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